European Forest Carbon, Wood-Use, and Food-Land Trade-offs Under Constrained Regional Policy Design

Application brief for the Chalmers doctoral position “Land Use and Climate Impacts”

This brief summarizes a portfolio-scale research project built to demonstrate direct fit for the Chalmers doctoral position. The repository combines official European and FAO data, transparent carbon-accounting logic, subnational spatial analysis, and a small constrained allocation model to study trade-offs between forest carbon, wood use, food-related land capacity, and policy design.

RESEARCH QUESTION

How do alternative forest-biomass strategies across Europe compare when climate performance is evaluated through forest carbon retention, substitution effects, and biomass use, and how do those forestry choices interact with agricultural land that may be important for food-related production capacity?

SCENARIO DESIGN

- Conservation Priority: reduce total harvest by 15%, taking the cut from wood fuel first.
- Material Cascade: keep harvest constant but shift 20% of wood fuel volume into industrial roundwood uses.
- Bioenergy Push: increase total harvest by 15% and allocate the additional harvest to direct energy use.
- Food-Land Safeguard: assign climate value to retaining a marginal share of agricultural land against further biomass diversion.

DATA AND SCOPE

- FAOSTAT land-use statistics for forest area and living biomass carbon stock.
- FAOSTAT forestry statistics for roundwood, industrial roundwood, wood fuel, and sawnwood production.
- Eurostat GISCO country and NUTS-2 geometry.
- Eurostat regional land-cover data for woodland, cropland, grassland, fodder, and permanent crops.
- Eurostat crop-production data for selected food crop area, production, and yield.
- Spatial domain: EU-27 countries and NUTS-2 regions. Analysis year: 2024 for country data and 2022 for the selected regional land-cover layer.

WHY THIS MATTERS FOR THE POSITION

- Shows land-use and biomass systems thinking rather than generic climate visualization.
- Demonstrates carbon-accounting logic and scientific computing in Python.
- Uses European and subnational spatial data directly relevant to the advertised research scope.
- Adds a simple biophysical-economic allocation step instead of stopping at descriptive scenario comparison.

Methods and Main Findings

METHODS

The country-level model estimates scenario performance through forest carbon retention, substitution gain, and supply-chain savings. The forest carbon opportunity-cost term is country-specific and increases where observed harvest intensity and carbon density are relatively high. A 400-draw sensitivity analysis is used to distinguish robust from assumption-sensitive results. At NUTS-2 level, country scenario values are downscaled by regional woodland share, and the agricultural side is strengthened with an official crop-production-intensity index derived from Eurostat data on cereals, pulses, potatoes, and vegetables.

The metric basis also differs across regional layers. The Sweden empirical rerun is a forest-only comparison reported in EUR per forest hectare, while the integrated and optimized NUTS-2 policy layers compare forest and food-land options together in EUR per hectare of total land. That distinction explains why SE22 can keep Material Cascade as its best forest option while still selecting Food-Land Safeguard in the integrated screen.

The final extension adds a small mixed-integer allocation model that chooses exactly one option per NUTS-2 region while preserving at least 97% of baseline regionalized biomass supply and safeguarding at least 22% of available crop-capacity proxy. This is still a screening model rather than a sector-equilibrium model, but it moves the project closer to the style of biophysical-economic land-use analysis described in the Chalmers call.

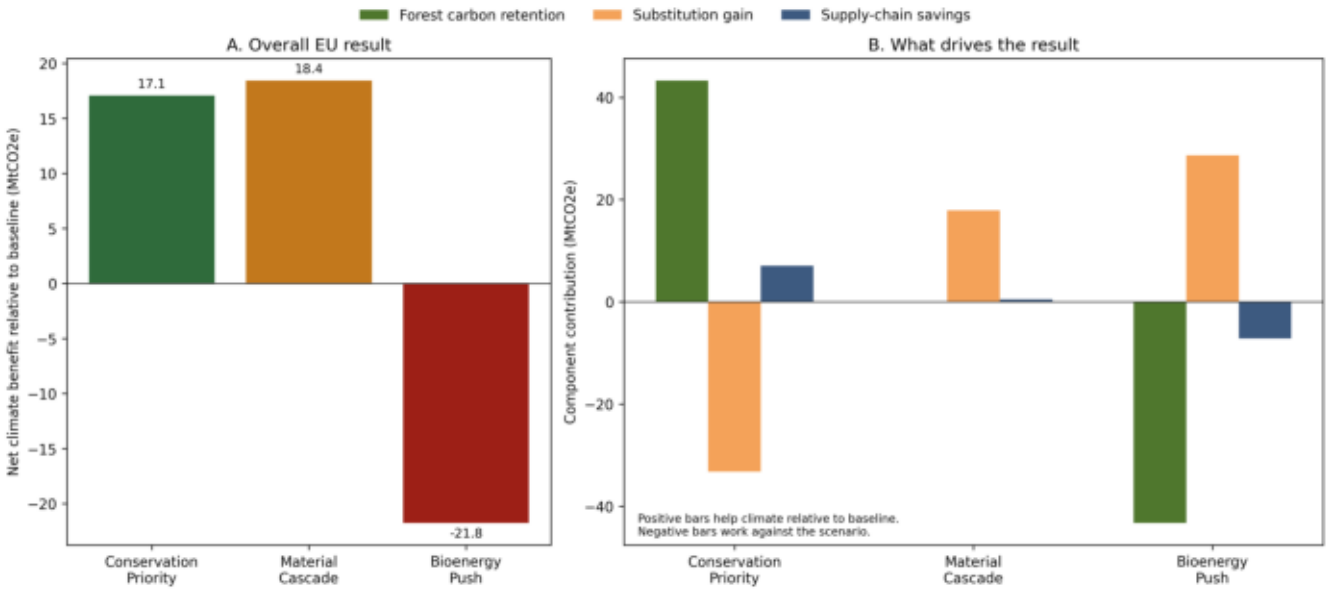
MAIN FINDINGS

- Material Cascade: about 18.4 $MtCO_2e$ net climate benefit relative to baseline.
- Conservation Priority: about 17.1 $MtCO_2e$.
- Bioenergy Push: about -21.8 $MtCO_2e$.
- Across 400 sensitivity runs, Bioenergy Push never becomes the best EU-wide option.
- Unconstrained integrated NUTS-2 screen: 92 Material Cascade, 79 Conservation Priority, 40 Food-Land Safeguard, 0 Bioenergy Push.
- Constrained regional portfolio: 123 Material Cascade, 34 Conservation Priority, 54 Food-Land Safeguard, 0 Bioenergy Push.
- 45 regions switch away from their unconstrained locally best option once the Europe-wide safeguards are enforced.
- The optimized portfolio retains about 97.0% of baseline regionalized biomass supply and safeguards about 45.6% of available crop-capacity proxy.
- Total optimized portfolio value at EUR 100/tCO_2: about EUR 2.15 billion.

INTERPRETATION

The project’s value lies in combining forest carbon, material substitution, biomass extraction, food-land competition, subnational spatial analysis, and constrained policy selection in one transparent workflow. It shows the ability to formulate the right kind of research question and to be explicit about what the model does and does not capture.

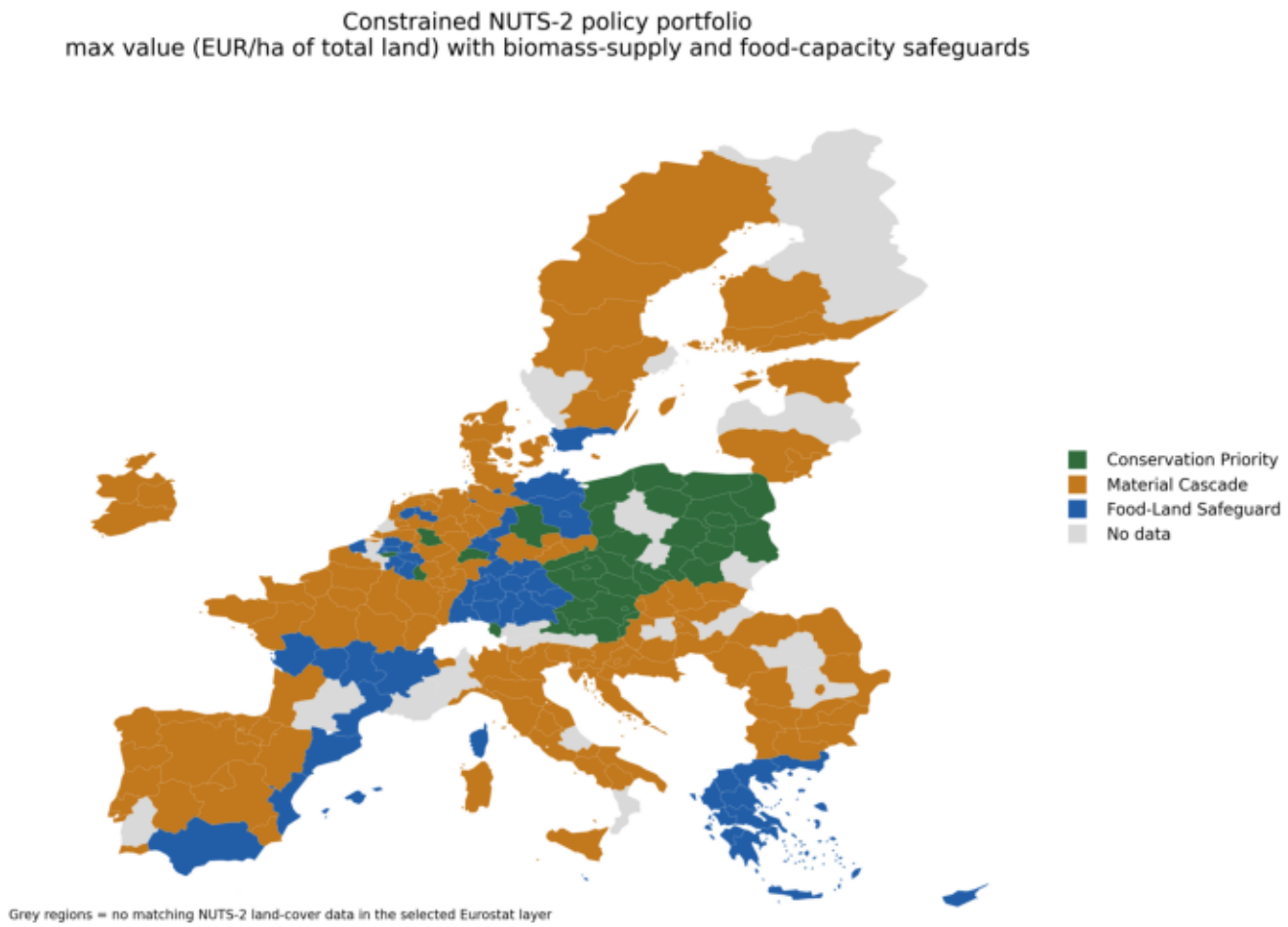
Figure 1. EU Scenario Decomposition



The EU-level decomposition figure is the clearest entry point for a committee reader. It shows that both material cascade and conservation outperform bioenergy expansion, but for different reasons: material use gains more through substitution, while conservation gains more through avoided forest carbon loss.

This figure is the best single summary of the project’s carbon-accounting logic.

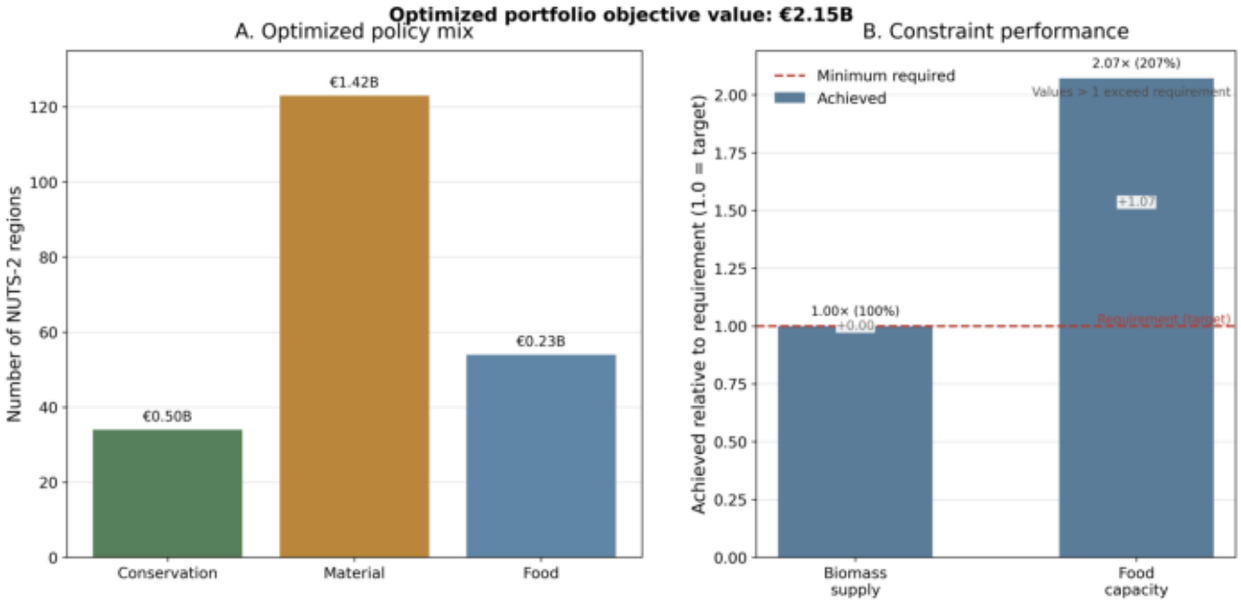
Figure 2. Constrained NUTS-2 Policy Map



The constrained NUTS-2 map demonstrates subnational spatial analysis and policy trade-off reasoning. It shows that once biomass supply and food-capacity safeguards are imposed at EU scale, the spatial portfolio shifts away from a purely local winner-takes-all ranking.

This figure most directly reflects the European and sub-national trade-off language in the Chalmers advertisement.

Figure 3. Constrained Portfolio Summary and PhD Extension Path



The constrained portfolio summary shows that the project is not purely descriptive. It selects a regional policy mix under explicit biomass-supply and crop-capacity safeguards, which is much closer to the advertised need for biophysical and economic modeling.

Most credible PhD extensions from this base: better subnational forestry data, direct regional agricultural production or GHG indicators, biodiversity constraints, tighter literature calibration, and a richer biophysical-economic allocation model.